Gesare-n / Movie-Recommendation-System

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README

Production-Grade Movie Recommendation System

Project Overview

This project implements an industry-standard movie recommendation engine designed to enhance user engagement and retention on digital entertainment platforms. By leveraging advanced supervised modeling techniques, including Collaborative Filtering (SVD), Content-Based Filtering, and Neural Collaborative Filtering (NCF), the system provides highly personalized movie suggestions that address the "choice paralysis" and "cold start" challenges common in the streaming industry.

Key Features

- **Hybrid Recommendation Engine**: Combines the behavioral insights of Collaborative Filtering with the attribute-based precision of Content-Based Filtering.
- **Advanced Evaluation Metrics**: Utilizes NDCG@10 (Normalized Discounted Cumulative Gain) and Precision@K to measure ranking effectiveness, moving beyond simple rating accuracy (RMSE).
- **Temporal Validation Strategy**: Implements a time-based train-test split to ensure model performance on future, unseen data, simulating a real-world production environment.
- **Deep Learning Baseline**: Introduces Neural Collaborative Filtering (NCF) using PyTorch to capture complex, non-linear user-item interactions.
- **Cold Start Solution**: Employs movie metadata (genres and tags) to provide intelligent recommendations for new content and users with sparse history.

Repository Structure

```
.
├── Industry_Standard_Recommendation_System.ipynb  # Main project notebook with analysis and modeling
├── ml-latest-small/                               # Dataset directory (MovieLens 100k)
│   ├── links.csv
│   ├── movies.csv
│   ├── ratings.csv
```

```
│   ├── tags.csv
│   └── README.txt
└── README.md                           # Project documentation
```

Getting Started

Prerequisites

Ensure you have the following Python packages installed:

- `pandas`
- `numpy`
- `scikit-learn`
- `scikit-surprise`
- `torch`

Installation

1. Clone the repository:

```
git clone https://github.com/your-username/Movie-Recommendation-System.git
cd Movie-Recommendation-System
```

2. Install dependencies:

```
pip install pandas numpy scikit-learn scikit-surprise torch
```

Running the Project

Open the `Industry_Standard_Recommendation_System.ipynb` notebook in a Jupyter environment to view the full analysis, modeling process, and evaluation results.

Data Source

The project uses the [MovieLens ml-latest-small dataset], which contains 100,836 ratings and 3,683 tag applications across 9,742 movies by 610 users.

Evaluation Results

Metric	SVD Model Performance
RMSE	0.8775
NDCG@10	0.8558

Business Impact

- **Increased Engagement**: Personalized discovery leads to longer user sessions.
- **Reduced Churn**: Relevant recommendations improve subscriber satisfaction and retention.
- **Content ROI**: Ensures new and niche content is discoverable, maximizing the value of the entire library.

Future Roadmap

- **Real-time Optimization**: Implementing streaming data pipelines for instant profile updates.
- **Model Explainability**: Integrating tools like LIME to provide transparent "Why this was recommended" features.

- **Enterprise Scalability**: Transitioning to cloud-based Spark environments for global-scale deployment.

Releases

No releases published
Create a new release

Packages

No packages published
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Contributors 1

Gesare-n Nicole Gesare Onyiego

Languages

- Jupyter Notebook 100.0%